1 A trolley with a magnet attached is placed on a thin steel sheet as shown in Fig. 1.1.

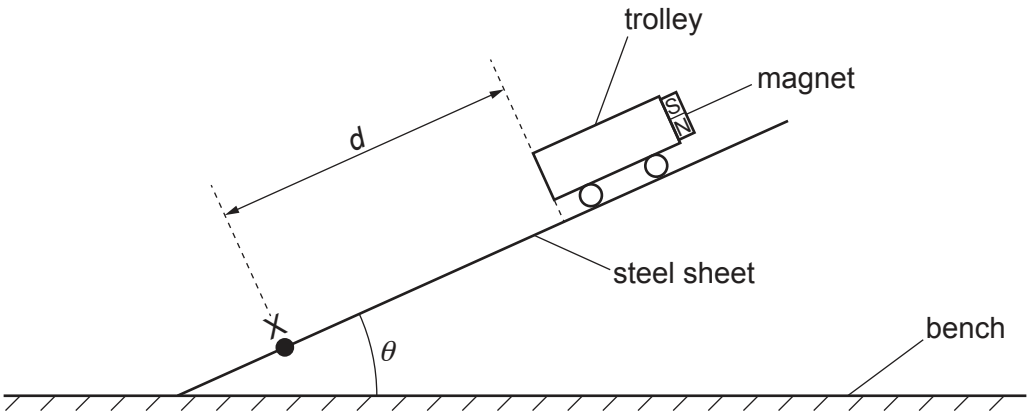

Fig. 1.1

The angle between the sheet and the bench is θ. The distance from point X to the trolley is d.

The trolley is released from rest and travels down the slope. The velocity v of the trolley at X is determined using a light gate.

It is suggested that v is related to θ by the relationship

$$mp \sin\theta - qB = \frac{mv^2}{2d}$$

where m is the mass of the trolley and magnet, B is the magnetic flux density between the magnet and the steel sheet, and p and q are constants.

Plan a laboratory experiment to test the relationship between v and θ.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for p and q.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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